Number-Theoretic π in the Casimir Effect: A Third Pathway in the Geometric Vocabulary

Version 2 — updated May 11, 2026. Adds Section 5.1 (prior mathematical results and the rationality observation).

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Abstract

A geometric vocabulary for reading the dimensionless factors in physics equations — where 2π denotes a circumference, √ a geometric mean, 4π a sphere surface, and D the spatial dimensionality — was previously applied to 55 equations spanning gravity, cosmology, thermodynamics, and oscillatory dynamics, and extended into quantum field theory through the one-loop Schwinger correction. The QFT extension identified two pathways by which π enters physical equations: geometric π (from angular integrals and sphere surfaces) and statistical π (from Gaussian integrals via $\Gamma(1/2) = \sqrt{\pi}$). This paper applies the vocabulary to the Casimir effect, $F/A = -\pi^2\hbar c/(240d^4)$, and identifies a third pathway: number-theoretic π, entering through Euler's formula for the Riemann zeta function at even arguments, $\zeta(2k) = |B_{2k}|(2\pi)^{2k}/[2(2k)!]$. The three pathways are shown to be observationally distinct via the D-test: geometric π appears for all spatial dimensions D, while number-theoretic π appears only when ζ(D+1) has an even argument (D odd) and vanishes entirely at odd arguments (D even), where ζ reduces to irrational constants with no known π dependence. The Casimir effect is the only equation in the 55-equation catalog where π enters through two different pathways simultaneously. An explicit boundary is identified where the geometric vocabulary ceases to apply: the Casimir force in an even number of spatial dimensions contains ζ(odd), which has no known geometric reading.

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1. Introduction

A geometric vocabulary for the dimensionless factors in physics equations was introduced in a previous work (Nguyen 2026a) and catalogued across 55 equations spanning gravitational, cosmological, thermodynamic, and oscillatory physics (Nguyen 2026b). The vocabulary reads:

- **2π** as the circumference of a physical boundary (horizon, orbit, phase cycle),
- **√** as a geometric mean bridging two competing physical regimes,
- **4π** as the surface area of a unit sphere (Gauss's law),
- **D** as the number of spatial dimensions,

and every dimensionless numerical factor in the catalog was shown to be one of these four geometric objects.

An extension into quantum field theory (Nguyen 2026c) traced the factors of π in the one-loop Schwinger correction $\alpha/(2\pi)$ to their geometric origins — phase circumferences $(2\pi)^D$, momentum-space sphere surfaces $S(S^3) = 2\pi^2$, and the electric flux surface 4π inside the fine structure constant α — and identified a distinction between two pathways by which π enters physics:

- **Type 1 — Geometric π**: from angular integrals over sphere surfaces and phase-space boundaries. Present in Coulomb's law, the Schwinger correction, and all Gauss's law equations.
- **Type 2 — Statistical π**: from Gaussian integrals, entering through $\Gamma(1/2) = \sqrt{\pi}$. Present in partition functions and path integrals. Flagged as the kill switch for extending the vocabulary to higher-loop QED corrections.

This paper applies the vocabulary to the Casimir effect (Equation #50 in the catalog) and reports three results:

1. A complete factor-by-factor decomposition of $F/A = -\pi^2\hbar c/(240d^4)$, verified by generalization to D spatial dimensions.
2. The identification of a third pathway — **number-theoretic π** — entering through Euler's formula for $\zeta(2k)$, which is observationally distinct from geometric π via the D-test.
3. An explicit boundary where the vocabulary stops: the Casimir effect in an even number of spatial dimensions contains $\zeta(\text{odd})$, which has no known expression involving π and therefore no geometric reading.

All identifications are classified as Standard, Consistent, New, or Speculative following the convention established in the QFT extension.

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2. Factor Decomposition of the Casimir Force

2.1 The Standard Result

The Casimir force per unit area between two parallel, perfectly conducting plates separated by a distance d in three spatial dimensions is (Casimir 1948):

$$F/A = -\pi^2\hbar c/(240\, d^4)$$

This result has been experimentally verified to the percent level (Lamoreaux 1997, Bressi et al. 2002).

2.2 From Energy to Force

The Casimir energy per unit area for a single massless scalar field with Dirichlet boundary conditions in D spatial dimensions is (Ambjørn and Wolfram 1983, Elizalde 1995):

$E/A = -[\Gamma((D+1)/2) \times \zeta(D+1)] / [2\text{^}(D+1) \times \pi\text{^}((D+1)/2)] \times \hbar c/d\text{^}D$

For an electromagnetic field, the energy acquires a factor of (D−1) for the transverse polarization count, and the force acquires an additional factor of D from differentiation with respect to the plate separation:

$F/A = -D(D-1) \times [\Gamma((D+1)/2) \times \zeta(D+1)] / [2\text{^}(D+1) \times \pi\text{^}((D+1)/2)] \times \hbar c/d\text{^}(D+1)$

At D = 3, this reduces to the standard result. The scalar energy coefficient is:

$C_3(\text{scalar}) = -\Gamma(2) \times \zeta(4) / [2^4 \times \pi^2] = -1 \times (\pi^4/90) / (16\pi^2) = -\pi^2/1440$

Multiplying by 2 polarizations gives $-\pi^2/720$ (energy), and differentiating gives $-\pi^2/240$ (force). Both are numerically verified.

2.3 Decomposition of the Factor 240

The number 240 is not arbitrary. It decomposes as:

- **1440** (scalar energy denominator) = $2 \times (2k)! \times |1/B_{2k}| \times 2\text{^}D$, from the Euler formula for ζ(4) combined with momentum-space measure factors
- **720** = 1440/2, where the 2 is the electromagnetic polarization count (D−1 = 2 at D = 3)
- **240** = 720/3, where the 3 is the spatial dimensionality D (from ∂/∂d of d^(−D))

This decomposition is unique: it is the only factorization of 240 in which every factor has independent D-dependence, a distinct physical origin, and produces the correct generalization at arbitrary D.

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3. Factor-by-Factor Vocabulary Reading

Each factor in the D-dimensional Casimir force is identified below, with its classification.

3.1 ℏc/d^(D+1) — Dimensional Backbone

The quantum-relativistic energy scale at separation d, setting the dimensions of force per area. Not geometric — this is the dimensional backbone of the equation.

Classification: Standard.

3.2 D — Spatial Dimensionality

From ∂/∂d of d^(−D). This is the number of spatial dimensions, the same "word" that appears in

$v\_rms = \sqrt{(Dk\_BT/m)}$ (Equation #53) and the Debye frequency (Equation #55). It generalizes correctly: in $D = 2$ spatial dimensions, the force has a $d^{(-3)}$ dependence; in $D = 5$, a $d^{(-6)}$ dependence.

Classification: Standard. D-test: pass.

3.3 (D−1) — Polarization Degeneracy

The number of independent transverse polarizations of a massless vector field in D spatial dimensions. In D + 1 spacetime dimensions, a massless spin-1 field has D + 1 components; subtracting one for gauge freedom and one for the constraint equation gives D − 1 physical degrees of freedom.

This is a counting argument, not geometry. At $D = 3$, it equals 2 — the same non-geometric factor identified in the Planck radiation law (Equation #48 in the catalog). In $D = 2$, it reduces to 1 polarization, consistent with known results for 2+1-dimensional electrodynamics. In $D = 1$, it gives 0, correctly indicating that there are no transverse modes in 1+1 dimensions.

Classification: Standard. Not geometric (degeneracy count). D-test: pass.

3.4 Γ((D+1)/2) — Momentum-Space Sphere Surface

From the angular integral over the (D−1)-dimensional momentum sphere. This factor is part of the surface area $\Omega\_(D-1) = 2\pi^{((D-1)/2)}/\Gamma((D-1)/2)$, combined with the measure factors from converting the momentum integral to polar coordinates.

This is geometric in the vocabulary sense: it encodes the surface of the sphere in momentum space through which virtual modes propagate. At $D = 3$, $\Gamma(2) = 1$ (trivial), but at other D values it contributes nontrivially.

Classification: Standard. Geometric (surface). D-test: pass.

3.5 2^(D+1) and π^((D+1)/2) — Momentum-Space Measure

From the integration measure $d^{(D-1)}k/(2\pi)^{(D-1)}$. The factors of 2π in the denominator are the circumferences of the phase-space cells in each momentum direction — the same phase circumferences that appear in Equations #41–45 in the catalog.

Classification: Standard. Geometric (phase circumference). D-test: pass.

3.6 ζ(D+1) — Mode Summation

The Riemann zeta function evaluated at D + 1, arising from the regularized sum over discrete mode numbers between the plates: $\Sigma\_\{n=1\}^\infty n^{(-(D+1))}$. This factor encodes the spectral content of the boundary condition.

At $D = 3$, $\zeta(4) = \pi^4/90$. The π^4 in this expression does not come from angular integration (Type 1) or Gaussian integration (Type 2). It enters through Euler's 1740 formula:

$\zeta(2k) = |B_{2k}| \times (2\pi)^{(2k)} / [2 \times (2k)!]$

where B_{2k} is the 2k-th Bernoulli number. The $(2\pi)^{(2k)}$ in this formula is the (2k)-th power of the period of the cotangent function, which enters because $\zeta(s)$ is related to $\cot(\pi z)$ via the partial fraction decomposition of the generating function $t/(e^t - 1)$.

The factors in $\zeta(4) = (2\pi)^4/1440$ decompose as:

- **$(2\pi)^4$**: four powers of a circumference — the periodicity of the Bernoulli generating function $t/(e^t - 1)$
- **2**: normalization factor in Euler's formula (combinatorial)
- **$(2k)! = 4! = 24$**: from the Bernoulli formula (combinatorial)
- **$|1/B_4| = 30$**: the inverse of the 4th Bernoulli number (number-theoretic)

The (2π) entering through Euler's formula can be read as the circumference of the thermal circle — the KMS periodicity of the Bose-Einstein distribution, which shares the same generating function $t/(e^t - 1)$. This identification connects the mode-counting (2π) to the thermal (2π) that generates Hawking and Unruh temperatures in Equations #1–3 of the catalog.

Classification: The $\zeta(D+1)$ factor itself is Standard. The identification of (2π) inside Euler's formula as a circumference (thermal circle / KMS periodicity) is Speculative — it is an interpretation, not a derivation. The decomposition of 1440 into geometric and combinatorial factors is Consistent.

D-test: pass — $\zeta(D+1)$ generalizes correctly to all D.

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4. Three Pathways for π

4.1 The Classification

The Casimir analysis, combined with the results of the QFT extension, yields three observationally distinguishable pathways by which π enters physics equations:

Pathway 1 — Geometric π. From angular integrals over sphere surfaces, phase-space boundaries, and physical circumferences. Mathematically: $\int d\Omega$ over $S^{(D-1)} \to \Omega_{(D-1)} \to \pi^{((D-1)/2)}$. Present in Coulomb's law, the Schwinger correction, all Gauss's law equations, and the momentum-space measure of the Casimir calculation.

Pathway 2 — Statistical π. From Gaussian integrals, entering through $\Gamma(1/2) = \sqrt{\pi}$. Mathematically: $\int \exp(-x^2)dx = \sqrt{\pi}$. Present in partition functions and path integrals. Not present in the Casimir effect (no Gaussian integration appears in the mode-summation derivation). Flagged in the QFT extension as the kill switch for higher-loop corrections.

Pathway 3 — Number-theoretic π. From Euler's formula for $\zeta(2k)$, entering through the periodicity of the Bernoulli generating function. Mathematically: $\zeta(2k) = |B_{2k}|(2\pi)^{(2k)}/[2(2k)!]$. Present in the Casimir effect (via $\zeta(4) = \pi^4/90$), the Stefan-Boltzmann law (via the same $\zeta(4)$), and black-body radiation.

4.2 Observational Distinctness: The D-Test

The three pathways are not merely notational variants. They are observationally distinguishable through their behavior under the D-test (generalization to D spatial dimensions):

- **Pathway 1** produces π for all values of D. The surface area $\Omega_{(D-1)}$ always contains $\pi^{((D-1)/2)}$, regardless of whether D is even or odd.
- **Pathway 2** produces $\sqrt{\pi}$ whenever a Gaussian integral is performed, independent of D.
- **Pathway 3** produces π only when $\zeta(D+1)$ has an even argument (i.e., D is odd). When D is even, $\zeta(D+1)$ evaluates to an irrational constant with no known closed form involving π.

This last point deserves emphasis. The values of the Riemann zeta function at odd positive integers — $\zeta(3)$ (Apéry's constant ≈ 1.202), $\zeta(5) \approx 1.037$, $\zeta(7) \approx 1.008$ — have no known expression as rational multiples of powers of π. Apéry (1978) proved $\zeta(3)$ is irrational; whether $\zeta(3)/\pi^3$ is rational remains an open problem in number theory. No closed form for any ζ(odd) in terms of π is known.

The consequence for the Casimir effect is:

Spatial dimensions D	ζ argument	ζ value	Contains π?
1	$\zeta(2) = \pi^2/6$	1.6449...	Yes (Type 3)
2	$\zeta(3)$	1.2021...	No
3	$\zeta(4) = \pi^4/90$	1.0823...	Yes (Type 3)
4	$\zeta(5)$	1.0369...	No
5	$\zeta(6) = \pi^6/945$	1.0173...	Yes (Type 3)
6	$\zeta(7)$	1.0083...	No

The on/off pattern is sharp: Type 3 π is present at odd D and absent at even D, with no gradual transition. This behavioral difference is the empirical proof that Pathway 3 is distinct from Pathway 1, which produces π at every D without exception.

4.3 The Casimir Effect as a Two-Pathway Equation

The Casimir force at D = 3 contains π through two simultaneous pathways:

- **Pathway 1**: the $(2\pi)^2$ from the momentum-space measure $d^2k/(2\pi)^2$, encoding the phase circumferences of the transverse momentum directions.
- **Pathway 3**: the $(2\pi)^4$ inside $\zeta(4) = (2\pi)^4/1440$, encoding the periodicity of the Bernoulli generating function.

In standard notation, both appear as "π^2" in the final result, and the two origins are invisible. The geometric vocabulary distinguishes them. This is the only equation in the 55-equation catalog where the same symbol π enters through two different pathways in the same formula.

4.4 Cross-Reference Table

The three-pathway classification applied across the catalog and its extensions:

Equation	Pathway 1 (Geometric)	Pathway 2 (Statistical)	Pathway 3 (Number-theoretic)
Coulomb's law (#46)	✓ (4π flux surface)	—	—
Gauss's law (#8–10)	✓ (4π flux surface)	—	—
Phase equations (#41–45)	✓ (2π phase circle)	—	—
Schwinger correction	✓ (momentum sphere, phase)	—	—
Casimir effect (#50)	✓ (momentum measure)	—	✓ ($\zeta(4)$)
Stefan-Boltzmann law	✓ (angular integration)	—	✓ ($\zeta(4)$)
Planck radiation (#48)	✓ (phase, via $h = 2\pi\hbar$)	—	✓ ($\zeta(4)$ in integrated form)
Thermal de Broglie (#51)	✓ (phase circumference)	?	—
Higher-loop QED	✓ (momentum sphere)	✓ ($\Gamma(1/2) = \sqrt{\pi}$)	?

The question marks indicate open classifications that require further analysis.

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5. The Even/Odd Boundary: Where the Vocabulary Stops

The geometric vocabulary was designed to read the dimensionless factors in physics equations as circumferences, surfaces, bridges, and dimensionality counts. It applies successfully across 55 equations in D = 3 spatial dimensions.

The D-dimensional Casimir effect reveals an explicit boundary:

In an odd number of spatial dimensions (D = 1, 3, 5, ...), the Casimir force coefficient decomposes fully into geometric factors: sphere surfaces ($\Gamma((D+1)/2)$), phase circumferences ($2^{(D+1)}$, $\pi^{((D+1)/2)}$), dimensionality (D), polarization counts (D−1), and number-theoretic circumferences (ζ(even) containing $(2\pi)^{(2k)}$).

In an even number of spatial dimensions (D = 2, 4, 6, ...), the force coefficient contains ζ(odd) — a number-theoretic factor with no known expression involving π and therefore no known geometric reading in the vocabulary.

This is not a failure of the vocabulary at D = 3 (our universe has odd D, and the vocabulary applies). It is the identification of a sharp boundary beyond which the vocabulary cannot currently reach. Whether this boundary reflects a genuine structural difference between odd-D and even-D physics, or merely the current state of knowledge about ζ(odd), is an open question that connects the geometric vocabulary to one of the oldest unsolved problems in pure mathematics.

5.1 Prior Mathematical Results and the Rationality Observation

The mathematical fact underlying the even/odd boundary is established in the existing literature. Cheng et al. (2010) state explicitly: "In the case of odd-dimensional space, the Casimir energy density can be expressed by the Bernoulli numbers, while in the even case it can be expressed by the ζ-function." Dolan and Nash (1992) showed that the Casimir energy on spheres vanishes in all even dimensions but is nonzero (and alternates in sign) in odd dimensions. The connection between Casimir calculations and the heat kernel expansion, which in turn connects to the Atiyah-Singer index theorem through the same Bernoulli numbers, is reviewed in Vassilevich (2003).

Stated in terms of the vocabulary, the mathematical content is: at odd D, the scalar Casimir energy coefficient is a rational multiple of $\pi^{((D+1)/2)}$, with the rational part controlled by Bernoulli numbers:

$$C_D(\text{scalar, odd } D) = -\Gamma((D+1)/2) \times |B_{(D+1)}| / [2 \times (D+1)!] \times \pi^{((D+1)/2)}$$

The rationality sequence is:

D	C_D / π^((D+1)/2)	Rational form
1	−0.04167	−1/24
3	−0.000694	−1/1440
5	−0.0000331	−1/30240
7	−0.00000248	−1/403200

At even D, the coefficient involves ζ(odd) and is irrational — no expression as a rational multiple of any power of π is known.

The present contribution is not the mathematical fact, which is standard, but the vocabulary reading: this even/odd split corresponds precisely to the boundary between equations whose dimensionless factors admit a complete geometric interpretation (circumferences, surfaces, rational combinatorics) and equations that contain factors with no known geometric reading. The three-pathway classification of π, the cross-reference table (Section 4.4), and the identification of the Casimir effect as a two-pathway equation (Section 4.3) are new to this work.

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6. Kill Switches

Two independent kill switches constrain the three-pathway classification:

Kill switch 1 (from the QFT extension): Statistical π at higher loops. If higher-loop QED corrections introduce $\Gamma(1/2) = \sqrt{\pi}$ in a way that resists geometric interpretation, Pathway 2 becomes a genuine obstacle for the vocabulary. This remains the primary threat identified in the QFT extension and is unchanged by the present analysis.

Kill switch 2 (new): Closed form for ζ(odd) involving π. If a closed form for ζ(3), ζ(5), or any ζ(odd) is discovered that expresses it as a rational multiple of a power of π, the behavioral distinction between Pathways 1 and 3 collapses. The even/odd boundary would dissolve, and the three-pathway classification would reduce to two. No such closed form is currently known, and extensive computational searches have found no evidence for one. The classification is consistent with the present state of number theory but is contingent on it.

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7. Summary

The Casimir effect, read through the geometric vocabulary, yields three results:

1. The factor 240 decomposes uniquely as (D−1) polarizations × D dimensionality × mode-counting factors from ζ(D+1), verified by generalization to arbitrary D.
2. π enters the Casimir force through two simultaneous pathways — geometric (momentum-space measure) and number-theoretic (Euler's formula for ζ(4)) — making it the only equation in the catalog where a single symbol carries two distinguishable geometric meanings.
3. The geometric vocabulary applies fully to the Casimir effect in odd spatial dimensions but encounters an explicit boundary at even dimensions, where ζ(odd) introduces factors with no known geometric reading.

The three-pathway classification (geometric, statistical, number-theoretic) is an observational classification based on behavioral differences under the D-test. Whether it reflects three fundamentally different mechanisms or one mechanism filtered through three mathematical structures is an open question.

Changelog (v2)

- **Added Section 5.1**: Prior mathematical results establishing the Bernoulli/ζ(odd) split across dimensions (Cheng et al. 2010, Dolan and Nash 1992, Vassilevich 2003). Explicit rationality table for odd-D coefficients. Clarifies that the mathematical content is standard; the contribution is the vocabulary reading and three-pathway classification.
- **Added references**: Cheng et al. (2010), Dolan and Nash (1992), Vassilevich (2003).

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